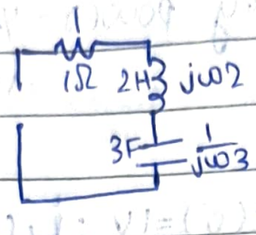
AC circuits

Recipe: (for steady state)

AC circuit; input frequency ω → "complex value" DC current
- sources
- elements

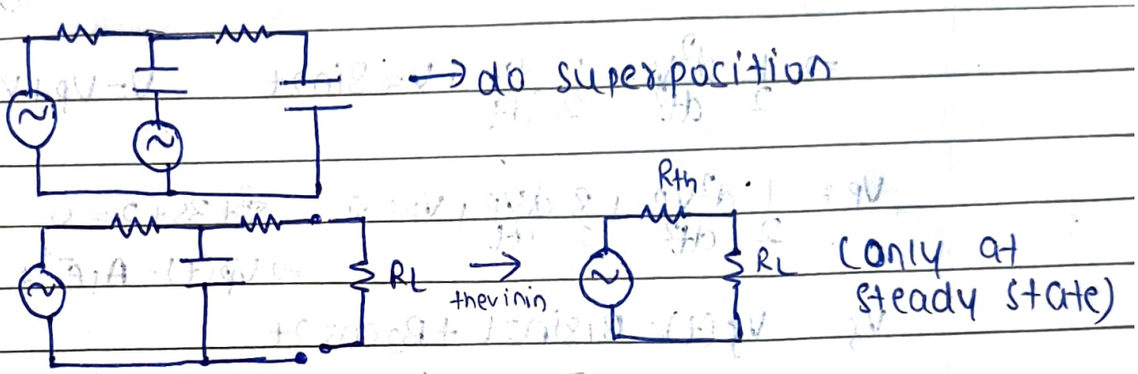
① L, C, R → $j\omega L$, $\frac{1}{j\omega C}$, R (No initial charge)



② Source $A_1\cos\omega t + A_2(-\sin\omega t)$ → $A_1 + jA_2$ complex DC voltage

③ Solve like normal resistive circuit and write it in a nice form $V_c(j\omega) = a + bj$; $a, b \in \mathbb{R}$

④ $v_c(t) = a\cos(\omega t) + b(-\sin(\omega t))$



(only at steady state)

Power: $P(t) = v(t)\,i(t)$

Average power: $P(t) = \frac{1}{T}\int_0^T v(t)\,i(t)\,dt$

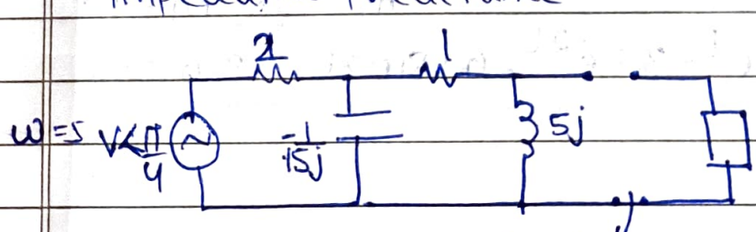
Quantity of interest: $\frac{(a_1 + a_2 j)(b_1 + b_2 j)\cdots}{(c_1 + c_2 j)\cdots}$

⇒ phase wrt $\cos(\omega t + \phi)$: $\phi = \tan^{-1}\left(\frac{a_2}{a_1}\right) + \tan^{-1}\left(\frac{b_2}{b_1}\right)\cdots - \tan^{-1}\left(\frac{c_2}{c_1}\right)\cdots$

Notations: $V_1\cos(\omega t) + V_2(-\sin\omega t) \Rightarrow V_1 + jV_2$

phasor notation: $V\angle\phi$ $\quad V^2 = V_1^2 + V_2^2$; $\phi = \tan^{-1}\frac{V_2}{V_1}$

impedance / reactance



Thevinin, norton

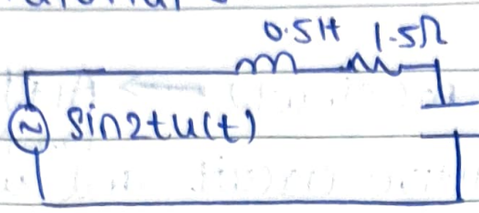
Thevinin impedance Z_{th} → $R_{th} + jX_{th}$ (resistance, reactance)

$Z_{th} = \left[(2 \,||\, (\frac{-1}{15j})) + 1\right] \,||\, 5j$

dependent source: to find R_{th} we add a source across the load with same AC freq.

$V = V_1 + V_2 j$ $\quad I = i_1 + j i_2$ → Avg power $= \frac{1}{2} Re\{VI^*\}$
$= (V_1 i_1 + V_2 i_2)/2$

Tutorial 8:

① 

$v_c(0^-) = 1V$; $i_L(0^-) = 2A$

$i_L(t) = \frac{d^2 v_c(t)}{dt^2}$

$i = \frac{dq}{dt} = C\frac{dv}{dt}$ $\quad q = CV$

$\rightarrow \frac{1}{2}\frac{di_L}{dt} + \frac{3i_L}{2} + v_c(t) = \sin 2t$

$\Rightarrow \frac{1}{2}\frac{d^2v}{dt^2} + \frac{3}{2}\frac{dv}{dt} + v = \sin 2t$ $\quad V = V_p + V_f$

V_p: $\frac{1}{2}\frac{d^2v_p}{dt^2} + \frac{3}{2}\frac{dv_p}{dt} + v_p = 0$ $\quad s^2 + 3s + 2 = 0$

$\Rightarrow v_p(t) = A_1 e^{-t} + A_2 e^{-2t}$

V_f: $v_f(t) = B_1 \sin 2t + B_2 \cos 2t$

$\Rightarrow \frac{1}{2}[-4B_1 \sin 2t - 4B_2 \cos 2t]$
$+ \frac{3}{2}[2B_1 \cos 2t - 2B_2 \sin 2t]$
$+ B_1 \sin 2t + B_2 \cos 2t = \sin 2t$

$\Rightarrow -2B_1 - 3B_2 + B_1 = 1 \Rightarrow B_1 + 3B_2 = -1$

$\Rightarrow -2B_2 + 3B_1 + B_2 = 0 \Rightarrow B_2 = 3B_1 \Rightarrow B_1 = -0.1$
$B_2 = -0.3$

$\Rightarrow v_c(t) = A_1 e^{-t} + A_2 e^{-2t} - 0.1 \sin 2t - 0.3 \cos 2t$

$\Rightarrow A_1 + A_2 - 0.3 = 1 \Rightarrow A_1 + A_2 = 1.3$

$i_L(t) \Rightarrow -A_1 e^{-t} - 2A_2 e^{-2t} - 0.2 \cos 2t + 0.6 \sin 2t$

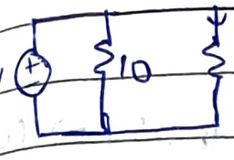
$\Rightarrow -A_1 - 2A_2 - 0.2 = 2 \Rightarrow A_1 + 2A_2 = -2.2$

$A_2 = -3.5$ $\quad A_1 = 4.8$

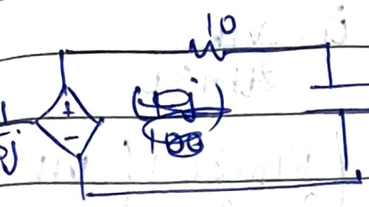
$$\boxed{v_c(t) = 4.8e^{-t} - 3.5e^{-2t} - 0.1\sin 2t - 0.3\cos 2t}$$

PageWork

D D M M Y Y Y Y

② 

$$\frac{(10+10j)\,1}{(10+10j)\,10j} = \frac{1+j}{100j} = 0.01(1-j)$$



$$\frac{-2(1-j)}{100}(-10j) = \frac{-j}{100}$$

$$\Rightarrow \frac{1}{100}\sin(100t)$$

$$i_L(j\omega) = \frac{1}{10j}$$

$$\frac{\left(\frac{1}{5j}\right)(-10-j)}{10-10j} = \frac{-1}{5(1-j)} = \frac{1}{j-1} = \frac{j+1}{-20}$$

$$\Rightarrow \frac{-1}{10}\cos(100t) + \frac{1}{10}\sin(100t)$$

$P = \frac{1}{2} Re \frac{|V|^2}{R} = (|V|/\sqrt{2})^2/R$ $\quad |V|/\sqrt{2} \rightarrow V_{DC} \rightarrow$ effective potential for Sinusoidal

$i_{eff} = |I|/\sqrt{2}$

$v(t) = V_m \cos(\omega t + \phi)$

Root mean square voltage

$$V_{RMS} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt} = \frac{|V_m|}{\sqrt{2}}$$

Compare V_{rms} & V_{eff} for Sinusoidal → $V_{rms} = V_{eff}$

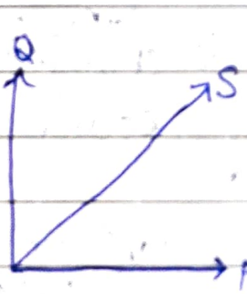
Complex power (S):

$$S = \frac{1}{2} V i^* = \frac{1}{2} Re(V i^*) + \frac{1}{2} Im(V i^*)$$

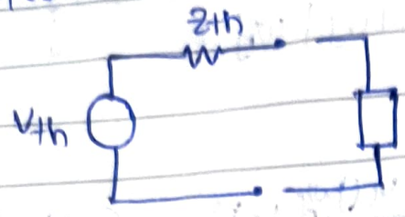
↓ $P_{avg} = P$ ↓ Q (Reactive Power)

reactive power: flows from source to reactive elements ⇄ back
↳ L & C.

Power triangle:



Maximum Power Transfer:



$V_{th}, Z_{th}, Z_L \rightarrow$ complex

$$i_L = \frac{V_{th}}{Z_{th} + Z_L}$$

$$P_{avg} = \frac{1}{2} Re\left[(V_L \cdot i_L^*) \cdot \left(\frac{V_{th} \cdot Z_L}{Z_{th} + Z_L}\right)\left(\frac{V_{th}}{Z_{th} + Z_L}\right)^* \right]$$

$$= \frac{1}{2} Re\left(\frac{|V_{th}|^2 \, Z_L}{|Z_L + Z_{th}|^2} \right)$$

$Z_{th} = R_{th} + jX_{th}$

$Z_L = R_L + jX_L$

$$(arg)\max_{Z_L} \frac{1}{2} Re\left\{ \frac{|V_{th}|^2 (R_L + jX_L)}{(R_L + R_{th})^2 + (X_L + X_{th})^2} \right\}$$

$$\Rightarrow \frac{|V_{th}|^2}{2} \arg\max_{R_L, X_L} Re\left(\frac{R_L + jX_L}{(R_L + R_{th})^2 + (X_L + X_{th})^2} \right)$$

$$= \frac{|V_{th}|^2}{2} \max_{R_L, X_L} \frac{R_L}{(R_L + R_{th})^2 + (X_L + X_{th})^2}$$

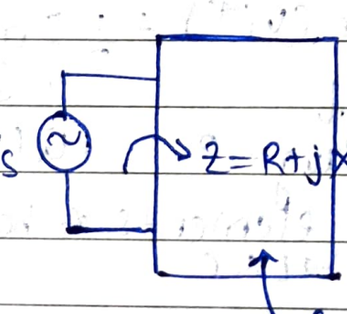
$$= \frac{|V_{th}|^2}{2} \max_{R_L} \left\{ \max_{X_L} \frac{R_L}{(R_L + R_{th})^2 + (X_L + X_{th})^2} \right\}$$

$$= \frac{|V_{th}|^2}{2} \max_{R_L} \left\{ \frac{R_L}{(R_L + R_{th})^2} \right\} \qquad \boxed{X_L = -X_{th}}$$

$$P_{max} = \frac{|V_{th}|^2}{8 R_{th}} \qquad \boxed{R_L = R_{th}}$$

$$\Rightarrow \boxed{Z_L = Z_{th}^*}$$

variable (ω)



If ω is changed, only X changes. We need to see avg power vs ω.

$$P = \frac{1}{2} Re\left\{ V_s^* \frac{V_s}{R + jX} \right\} = \frac{|V_s|^2}{2} Re\left\{ \frac{R}{R - jX} \right\} = \frac{|V_s|^2 R}{2(R^2 + X^2)}$$

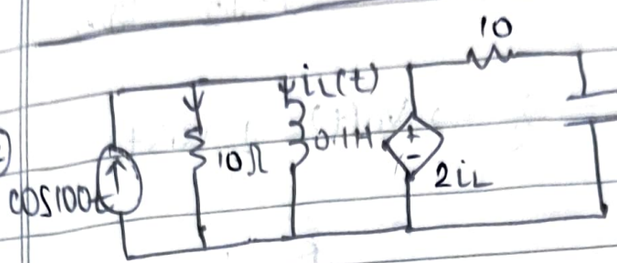
maximized when $X = 0 \Rightarrow$ {ω : X(ω) = 0}

$X = 0$ ⇔ V_s & i are in phase ($\Delta\phi = 0$)

↓

Resonance

ω for which it happens ⇒ resonant frequency

② 

$$10(\cos 100t - i_L(t)) = 0.1\frac{di_L(t)}{dt}$$

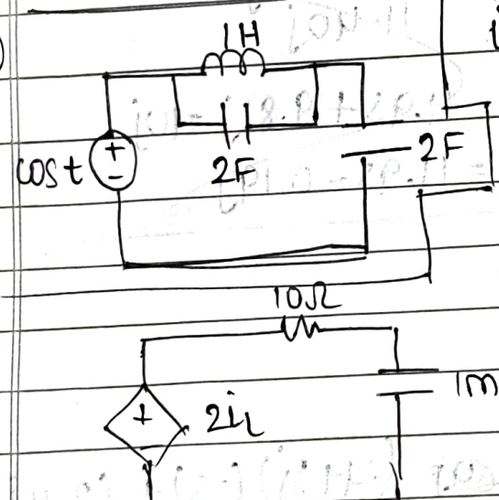
$$\Rightarrow \frac{di_L}{dt} + 100 i_L = 100\cos(100t)$$

$$d(e^{+100t} i_L) = 100\, e^{+100t}\cos(100t)$$

$$\Rightarrow i_L e^{+100t}\Big|_{-\infty}^{t} = \int_{-\infty}^{t} e^{100t}\cos(100t)\cdot 100\, dt$$

$$\Rightarrow i_L e^{100t} = \frac{e^{100t}}{2\cdot 100^2}[100][\sin 100t - \cos 100t]\Big|_{-\infty}^{t}$$

$$i_L = \frac{100}{200}[\sin(100t) - \cos(100t)]$$

③ 

$$\frac{[\sin 100t - \cos 100t]}{100} = 10i + 9(100)$$

$$\Rightarrow \cos 100t + \sin 100t = 10\frac{di}{dt} + 1000\, i$$

$$\Rightarrow 10\, d(e^{100t} i) = e^{100t}(\cos 100t + \sin 100t)$$

$$\Rightarrow 10\, e^{100t} i = \frac{e^{100t}}{2\cdot 100}[2\cdot \sin 100t]$$

$$\Rightarrow i = \frac{1}{1000}\sin(100t)$$

$$V = \int_{-\infty}^{t} \frac{1}{1000}\sin(100t) = \frac{\cos(100t)}{100}$$

$1/1000$

③



$$\frac{j \cdot \frac{1}{2}j}{j+\frac{1}{2}j} + \frac{1}{2j}$$

$$\frac{1}{2} \quad \frac{2j}{-2+1} + \frac{1}{2j}$$

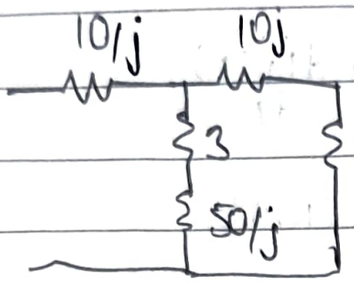
$$-j - j/2 = -3/2$$

$$I(j\omega) = \frac{2j}{3}$$

$$V_0(j) = \frac{1}{2j} \cdot \frac{2j}{3} = \frac{1}{3} \Rightarrow V_0(t) = \frac{1}{3}\cos t$$

---

④



$$f = \frac{1}{j \cdot 50 \times 2 \times 10^{-1}}$$

$$j \cdot 50 \cdot 0.2$$

$$10j+8$$

$$2j+3$$

$$(10j+8)(-50j+3)$$

$$\frac{(10j+8)(-2j+3) \cdot -10j}{8j+11}$$

$$11-40j$$

$$= \frac{500+24-400j+30j}{11-40j}$$

$$= \frac{596}{185} - \frac{2048}{185}j$$

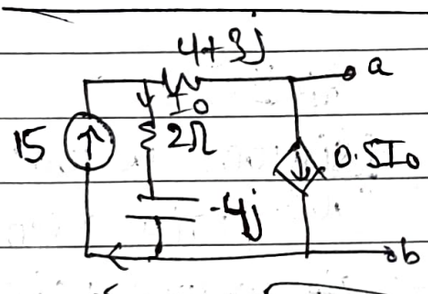
$$= \frac{524-370j}{11-40j}$$

$$= 11.95 + 9.81j - 10j$$

$$= 11.95 - 0.19j$$

---

⑤



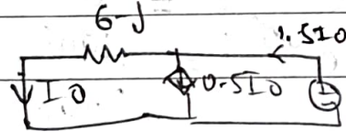
$I_0 = 10$

$$(4+3j)(-5) + (2-4j)(10)$$

$$= -20 - 15j + 20 - 40j$$

$$V_{th} = -55j$$

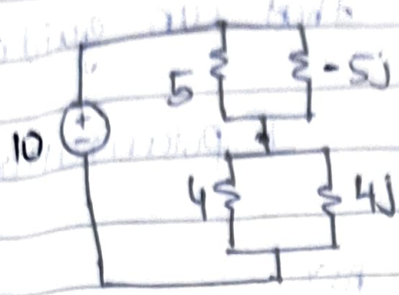
$Z_{th} =$



$$1 = (6-j)I_0 \Rightarrow I_0 = \frac{1}{6-j} \cdot \frac{6+j}{37}$$

$$\frac{1}{15I_0} \quad \frac{2}{3}(6+j) = 4+\frac{2}{3}j$$

⑥



$\frac{5(-5j)}{5-5j} + \frac{4(4j)}{4+4j}$

$= \frac{-5j}{1-j} + \frac{4j}{1+j}$

$= \frac{-5j(1+j)}{2} + \frac{4j(1-j)}{2}$

$i = \frac{20}{9-j}$ $\quad Z = \frac{-5j+5+4j+4}{2} = \frac{9-j}{2}$

$V_c = \left(\frac{20}{9-j}\right)\left(\frac{-5j(1+j)}{2}\right) = -50\left(\frac{j-1}{9-j}\right)$

$= 6.1 - 4.9j$

$= 7.8\cos(2t - 38.7°)$

$i = \left(\frac{20}{9-j}\right)\left(\frac{4j(1-j)}{2}\right)\frac{1}{4j}$ $\quad 10\left(\frac{1-j}{9-j}\right) = 1.22 - 0.98j$

$-1.56\cos(2t - 38.7°)$